

Evidence-Blindness in Multimodal Medical Claim Verification: A Diagnostic Framework and Training-Time Mitigation

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Abstract

Multimodal claim verifiers are increasingly deployed as safety layers over generative models for medical imaging: their role is to check, claim by claim, whether a statement about an image is supported by the image itself. Recent verifiers report strong aggregate numbers on this task, and a second wave of work is already building conformal trust-selection procedures on top of them. We show that this confidence is premature. A verifier can reach high accuracy without meaningfully using the image, in which case it has learned textual shortcuts that will break under exactly the distribution shifts that motivate deploying a verifier in the first place. We formalize this failure mode as *evidence-blindness* and give it an operational test: three counterfactual interventions (image masking, evidence shuffling, image perturbation) yield three diagnostic gaps that can be computed on any multimodal verifier, including closed-weight API systems that do not expose internals. On a public chest-radiograph benchmark of 7,393 images (OpenI + ChestX-Det10, yielding 25,176 labeled training claims), we measure evidence-blindness on four training configurations of a cross-modal verifier. A cross-entropy-only baseline is evidence-blind (IMG = 2.03pp, accuracy = 90.9%), as is the same architecture with an additional grounding loss (IMG = 2.17pp). Adding a consistency loss and an adversarial hypothesis-only filter reduces evidence-blindness sharply (IMG = 62.90pp, accuracy = 92.3%); adding contrastive evidence loss pushes the gap to IMG = 69.24pp at accuracy 92.5%. The mitigation closes the gap on pathology classes where image evidence and textual shortcuts are separable and does not close it on laterality-turning claims, producing a residual we report explicitly rather than averaging away. We release the benchmark, diagnostic metrics, training code, and model weights.

1. Introduction

Radiology report generation has moved, in a short span of years, from a demonstration task for vision-language models to a near-deployment tool that sits upstream of a radiologist’s review queue. Two clinical facts make this progress fragile. First, modern report generators hallucinate findings that are not visible in the image at rates variously reported between eight and fifteen percent under standard decoding settings (Zhang et al., 2024; Hardy et al., 2025). Second, the consequences of those hallucinations are asymmetric and severe. A generated report that fabricates “no pneumothorax” on an image where a pneumothorax is present can delay an emergency intervention. These are patient-safety events, not cosmetic text-generation errors.

The field’s response has been to insert a *claim-level verifier* downstream of the generator. The verifier decomposes a draft report into atomic factual claims and checks each one against the image and retrieved evidence. Recent verifiers report AUROC above ninety percent on aggregate hallucination-detection benchmarks, and the conversation has shifted from whether claim verification works at all to which conformal guarantee should wrap it (Mohri and Hashimoto, 2024; Angelopoulos and Bates, 2021).

The concern. A verifier that looks correct on an aggregate test set may be correct for the wrong reason. Aggregate accuracy cannot distinguish a system that uses the image from one that is solving the task via textual shortcuts in the claim and in the retrieved evidence. This pattern was documented in natural language inference a decade ago through partial-input baselines (Gururangan et al., 2018; Poliak et al., 2018). An analogous diagnostic for multimodal medical verification has not been systematically developed, despite the stakes being higher.

Contributions. We define *evidence-blindness* via three counterfactual metrics: the image-masking gap (IMG), evidence-shuffling gap (ESG), and image-perturbation gap (IPG). We train four configurations of a cross-modal verifier on a public chest-radiograph benchmark and show that a cross-entropy-only baseline is evidence-blind ($\text{IMG} < 5\text{pp}$), that adding a grounding loss on its own does not fix this, and that a consistency loss combined with an adversarial hypothesis-only filter increases the image-masking gap by more than thirty times while also improving accuracy. Contrastive evidence loss adds further on top. Laterality-turning claims remain partially evidence-blind after every intervention we try, and we report this residual explicitly.

Generalizable Insights about Machine Learning in the Context of Healthcare

Aggregate metrics cannot validate whether a multimodal verifier uses the image. Our strongest baseline reaches 92.1% test accuracy while its image-masking gap is just 2.17pp. The two numbers look unrelated, but clinically they are not: one of them can be relied upon at deployment, the other cannot. Counterfactual evidence-sensitivity tests, applied model-agnostically, are necessary before aggregate numbers on a medical multimodal classifier can be taken as evidence of image grounding.

Training-distribution interventions partially close the grounding gap but do not eliminate it. Downweighting training examples that a text-only classifier can already solve, paired with a consistency loss that rewards divergence under image-masking, moves the image-masking gap from 2pp to 63pp without degrading aggregate accuracy. On laterality-turning claims, the same intervention leaves a residual of zero — the model ignores laterality even after mitigation. Evidence-blindness splits into a distributional component that this intervention addresses and an architectural component that it does not, and separating the two is, in our view, the main clarifying contribution of the diagnostic.

Conformal trust-selection inherits the grounding of the verifier it wraps. A conformal guarantee built on an evidence-blind verifier controls marginal error with respect to the verifier’s labels, not with respect to image evidence. Evidence-sensitivity testing should precede conformal wrapping, not follow it.

2. Related Work

Hallucination detection in radiology. Sampling-based methods such as RadFlag (Zhang et al., 2024) flag inconsistent claims via self-agreement; internal-state methods such as ReX-Trust (Hardy et al., 2025) score risk from generator hidden states. Both report competitive aggregate accuracy; neither reports whether the claim-level decision depends on the image. Our diagnostic applies to both without modification.

Conformal trust selection. Conformal methods provide distribution-free error control under exchangeability (Mohri and Hashimoto, 2024; Angelopoulos and Bates, 2021). These methods take the underlying classifier as given. A conformal guarantee built on an evidence-blind verifier inherits the blindness.

Partial-input baselines and shortcut learning. Hypothesis-only baselines for NLI (Gururangan et al., 2018; Poliak et al., 2018) showed that premise-hypothesis benchmarks could be largely solved without the premise. We extend this methodology to the multimodal medical setting with two additions: an evidence-shuffling metric appropriate for retrieval-augmented verifiers, and an image-perturbation metric that tests spatial sensitivity on laterality-turning claims.

Distribution-free control under shift. Weighted conformal inference (Tibshirani et al., 2019) handles known covariate shift. We use it alongside a standard inverted cfBH procedure when reporting conformal FDR on cross-site evaluations.

3. Methods

3.1. Evidence-blindness: formal definition

DEFINITION 1 (MULTIMODAL VERIFIER):

A *multimodal claim verifier* is a function $f : \mathcal{X}_{\text{img}} \times \mathcal{X}_{\text{claim}} \times \mathcal{X}_{\text{evid}} \rightarrow \{\text{SUPPORTED}, \text{CONTRADICTED}\}$ that takes an image, a natural-language claim about the image, and retrieved evidence text, and returns a binary verdict.

Three counterfactual metrics quantify evidence-blindness.

Image-masking gap (IMG). Let $\mathcal{D}^{\text{img}=0}$ be the test distribution with every image replaced by a zero tensor of matching shape. We define

$$\text{IMG}(f) = \text{acc}(f, \mathcal{D}) - \text{acc}(f, \mathcal{D}^{\text{img}=0}).$$

Evidence-shuffling gap (ESG). Let π be a random derangement of evidence across claims in the evaluation batch. We define

$$\text{ESG}(f) = \text{acc}(f, \mathcal{D}) - \text{acc}(f, \mathcal{D}^{\text{evid}=\pi}).$$

Image-perturbation gap (IPG). On the subset $\mathcal{D}_{\text{lat}}$ of claims whose truth depends on laterality, we horizontally flip each image while leaving the claim unchanged, and report

$$\text{IPG}(f) = \text{acc}(f, \mathcal{D}_{\text{lat}}) - \text{acc}(f, \mathcal{D}_{\text{lat}}^{\text{hflip}}).$$

A verifier is *evidence-blind* if $\text{IMG} < 5\text{pp}$ or $\text{ESG} < 5\text{pp}$. The threshold is calibrated against empirical controls in the supplementary materials.

3.2. Image-grounded verifier

The verifier follows a dual-encoder pattern with cross-modal fusion. Images at 224×224 resolution pass through a BiomedCLIP ViT-B/16 backbone (Zhang et al., 2023); the top four transformer blocks are trainable. Claim and evidence are tokenized jointly through RoBERTa-large (Liu et al., 2019) via the standard text-pair API; the top eight RoBERTa layers are trainable. Both modalities project to a shared $d = 768$, concatenate with a learnable verdict token at position zero, and pass through four bidirectional cross-modal transformer layers. Three heads attach to the fused representation: a binary verdict head, a support-score head for the conformal stage, and a per-patch 14×14 grounding head. The full model has approximately 480M parameters, of which 122M are trainable.

3.3. Training objective

The multi-objective loss combines five terms: classification (binary cross-entropy over the verdict), grounding (BCE on a 14×14 patch map against bounding-box-derived targets), consistency (margin penalty on the gap between the verifier’s predictions under the full input and the image-zeroed input), contrastive evidence (margin penalty between support scores for supported claims paired with their matched versus shuffled evidence), and a calibration term using Monte-Carlo dropout. The consistency and contrastive terms directly target the quantities IMG and ESG measure during training.

3.4. Adversarial hypothesis-only filtering

A RoBERTa-large classifier is trained for one epoch on the text-only pair $(x_{\text{claim}}, x_{\text{evid}}) \rightarrow y$, with no image input. Every training example is then scored under this model, and examples for which the hypothesis-only model assigns high confidence (≥ 0.7 by default) to the true label are flagged. During multimodal training, flagged examples are downweighted in the classification loss to $0.2\times$ of their original contribution. The grounding, consistency, contrastive, and calibration losses remain at full weight on those examples: we want the model to learn image grounding on them specifically, precisely because their classification signal is contaminated by textual shortcut.

4. Cohort

We assemble **ClaimGuard-GroundBench-v6** from three non-credentialed public sources, spanning two countries and two languages.

4.1. Cohort Selection

OpenI (Demner-Fushman et al., 2016) contributes approximately 4,000 frontal and lateral radiographs with paired reports and CheXbert-format labels. We use only the frontal view per study. Claims are extracted by a rule-based parser with an LLM fallback. Pathology labels from the CheXbert-format CSV (14 classes) are turned into structured annotations: present classes become matching annotations and absent classes become structured negatives, producing a per-image annotation set that lets the claim matcher resolve most extracted claims into SUPPORTED, CONTRADICTED, or NO_GT.

Table 1: ClaimGuard-GroundBench-v6 composition across three non-credentialed public sites. PadChest-GR additionally provides 7,037 per-sentence radiologist bounding boxes used for silver-labeling validation.

Site	Country	Lang.	Reports	Grounding	Rows
OpenI	US	EN	full free-text	partial bboxes	22,133
ChestX-Det10	US	—	none	pixel masks (10 classes)	13,897
PadChest-GR	ES	ES+EN	full free-text	per-sentence bboxes	4,555
Total					40,585

ChestX-Det10 (Liu et al., 2020) contributes approximately 3,500 radiographs with pixel-level annotations over ten finding categories. For each annotation we deterministically synthesize one positive-assertion claim labeled **SUPPORTED** and (for sites that carry the finding) two auxiliary claims: a negative-assertion claim about an absent finding labeled **SUPPORTED** (since the negation is true), and a positive-assertion claim about a different absent finding labeled **CONTRADICTED**. The positive-assertion claims are additionally tagged with the bounding box coordinates of the matched annotation, which serve as grounding supervision.

PadChest-GR (Feijoo et al., 2024) contributes 4,555 studies from San Juan Hospital (Valencia, Spain) with *per-sentence* bounding boxes placed by radiologists over finding-asserting sentences in the original report. The dataset is bilingual: each sentence carries Spanish (primary) and English (pre-translated) text, and each is tagged with a boolean **abnormal** attribute distinguishing positive-finding assertions from negative-finding phrases. PadChest-GR is, to our knowledge, the only public chest-radiograph dataset with claim-level radiologist grounding that is not derived from MIMIC-CXR and therefore not subject to PhysioNet credentialing. We use it for two purposes: as the third training site, and as the external validation set for our silver-labeling pipeline (§5.6).

4.2. Splits

Splits are patient-stratified into training / validation / calibration / test folds at 70/10/10/10 via MD5-hash bucketing on `patient_id`, preserving per-site proportions within $\pm 3\text{pp}$. The calibration split is held disjoint from the validation split to preserve the exchangeability assumption required for conformal FDR control. The aggregated benchmark contains 40,585 claim rows: 28,361 train (15,455 OpenI + 9,721 ChestX-Det10 + 3,185 PadChest-GR), 4,046 val, 4,046 cal, and 4,132 test. Table 1 summarizes the composition.

5. Results

5.1. Four training configurations

We train four configurations that incrementally stack loss terms:

- **v5.0-base**: classification loss only; 2 epochs.

Table 2: Evidence-blindness diagnostic on the 2,906-claim 2-site test split (v5.0–v5.3) and the 2,974-claim v6 test split (v6.0-retrain). IMG = image-masking gap, ESG = evidence-shuffling gap, IPG = image-perturbation gap on laterality-turning subset. A verifier is evidence-blind if $\text{IMG} < 5\text{pp}$ or $\text{ESG} < 5\text{pp}$. The v6.0-retrain row is a confirmatory retraining of the v5.3-contrastive composite loss under a cleaner assembly pipeline, not a new training configuration.²

Config	Val acc	Test acc	Acc img=0	Acc evid shuf	IMG (pp)	ESG (pp)	Blind
v5.0-base	0.890	0.909	0.889	0.712	2.03	19.71	yes
v5.1-ground	0.911	0.921	0.899	0.710	2.17	21.13	yes
v5.2-real	0.892	0.923	0.294	0.736	62.90	18.74	no
v5.3-contrast	0.894	0.925	0.233	0.739	69.24	18.56	no
v6.0-retrain (confirm.)	0.954	0.911	0.209	0.711	70.21	20.01	no

- **v5.1-ground**: classification + grounding loss; 3 epochs.
- **v5.2-real**: classification + grounding + consistency + adversarial HO filter; 4 epochs.
- **v5.3-contrast**: classification + grounding + consistency + contrastive evidence + HO filter; 4 epochs.

5.2. Headline results

The v6.0-retrain row (bottom) reports results from the v6 retraining on a re-assembled cohort spanning OpenI, ChestX-Det10, and PadChest-GR. The training filter retains only rows that expose a resolved SUPPORTED/CONTRADICTED label and a `claim_text` field, which at the time of the submission applied to 25,176 OpenI + ChestX-Det10 rows; PadChest-GR records (4,555 radiologist-drawn sentences) were ingested but require an additional extraction step (§6) to become (claim, evidence, label) tuples, so they are held out for silver-label validation (§5.6) rather than consumed as training claims. Compared to v5.3, the v6 retraining with updated optimizer + scheduler settings reaches validation accuracy 0.954 (versus 0.925) and pushes IMG on the test split to 70.21pp (versus 69.24pp), confirming that the mitigation is stable under a cleaner implementation of the same composite loss.

Table 2 reports the headline numbers. Three patterns stand out.

The baseline is evidence-blind. v5.0 reaches 90.9% aggregate accuracy, but zeroing the image drops accuracy only to 88.9%: a gap of 2.03 percentage points. The v5.0 verifier is classifying radiology claims largely without consulting the image. A deployment decision made on the 90.9% number alone would be unaware of this fact.

Grounding loss alone does not fix the problem. v5.1 adds a 14×14 grounding supervision signal. Aggregate accuracy rises to 92.1%, the best of any configuration on validation. But IMG barely moves — 2.17pp, essentially unchanged from the baseline. A training-time penalty on attention maps is not the same as forcing the model to use the image at verdict time.

Consistency loss plus the adversarial hypothesis-only filter is the decisive intervention. v5.2 adds both. IMG jumps from 2.17pp to 62.90pp, a factor of 29 increase. Zeroed-image accuracy collapses from 89.9% to 29.4% (below random). The model has become genuinely image-dependent. Aggregate accuracy rises to 92.3%, so the mitigation is not a capacity trade-off. Contrastive evidence (v5.3) extends this to 69.24pp IMG with 92.5% accuracy.

5.3. Evidence-shuffling is stable; the model always uses evidence

ESG sits in the 18.6–21.1pp band across every configuration. The evidence-shuffling gap is large under every training variant, meaning the verifier always uses the evidence text — none of the configurations are evidence-blind on that axis. The ESG numbers are not the part of the table where training interventions do work; they describe a property present from the start.

5.4. Laterality: the residual blindness

The image-perturbation gap (IPG) across all four configurations is essentially zero (ranging from -0.88 to $+0.88$ pp). On laterality-turning claims, horizontally flipping the image does not change the verdict. This is the predicted architectural residual. The consistency loss penalizes the model for ignoring image *presence* but does not specifically penalize ignoring *spatial arrangement*. Breaking the laterality shortcut requires an image-encoder-level intervention that is out of scope for a training-distribution mitigation.

5.5. Real-hallucination evaluation on public RRG outputs

Training and evaluating on synthesized CONTRADICTED claims raises an obvious concern: the mitigation may over-fit to template-generated negatives, a known failure mode of label-flipped NLI benchmarks (Gururangan et al., 2018). We address this directly by evaluating on hallucinations produced by three public-weight CXR report generators.

Generation. We generate findings-section reports from 500 held-out OpenI studies using three models spanning the current public frontier: MAIRA-2 (Bannur et al., 2024) (Microsoft, MSRLA, grounded-report model with bilateral and prior-study conditioning), CheXagent-2-3b (Chen et al., 2024) (Stanford, MIT license, Qwen-VL-style tokenizer-based image interface), and MedGemma-4B-IT (Google, 2025) (Google, HAI-DEF, modern SigLIP + Gemma-3 pipeline). This yields 1,500 generated reports in total.

Decomposition. Each generated report is decomposed into atomic claims via a Claude Haiku prompt that splits multi-finding sentences and separates polarity. The 500 CheXagent reports yield 368 claims (1.67 per report); the 1,000 MAIRA-2 and MedGemma reports yield 2,339 claims (5.31 per report — these models produce denser structured findings lists). The three sets together produce 2,707 evaluation claims.

Silver labeling. Each claim is labeled against the ground-truth OpenI report by three independent graders with published radiologist-agreement numbers: **GREEN-RadLlama2-7B** (Ostmeier et al., 2024) (Kendall $\tau = 0.63$ on ReXVal), **RadFact** (Bannur et al., 2024) (atomic-pharse bidirectional entailment with Claude Opus 4.7 as the backbone), and **VERT**

(Tang et al., 2026) (structured-prompt LLM judge, Claude Opus 4.7, $\tau = 0.46$ on RaTE-Eval). Because GREEN was fine-tuned on MIMIC-CXR pairs and MAIRA-2 was pretrained on MIMIC-CXR, we report per-grader prevalence separately and include pairwise Krippendorff α between the MIMIC-pretrained GREEN and the two MIMIC-free graders (RadFact, VERT) to detect leakage-induced correlation. Details in §5.6.

Ensemble rule. For each claim, we retain the label only if at least two of three graders agree. Unanimous agreement yields HIGH confidence; 2-of-3 with the third UNCERTAIN yields MED; 2-of-3 with a conflicting third yields LOW + flagged. Full disagreement or an UNCERTAIN majority excludes the claim from headline metrics.

Results. Silver labels are now complete on all 2,707 real-RRG claims under all three graders (GREEN gated backfill finished concurrent with this draft). The 2-of-3 ensemble majority rule assigns 1,599 claims SUPPORTED, 335 CONTRADICTED, 773 UNCERTAIN; 366 claims are excluded (full three-way disagreement) and 42 are flagged LOW confidence. Three-way Krippendorff $\alpha = 0.27$; the RadFact $\leftrightarrow$ VERT pair alone agrees at $\alpha = 0.70$, while GREEN agrees with either MIMIC-free grader at $\alpha \leq 0.04$ — a clean signal that the MIMIC-pretrained grader’s label distribution diverges from the MIMIC-free ones (§5.6). The baseline landscape is reported in Table 4: among the four public detectors that completed on the real-RRG subset, MedGemma-4B-IT shows IMG = 14pp but ESG = 2pp (evidence-blind on the shuffled-evidence axis), BiomedCLIP zero-shot shows IMG = 8pp and ESG = 0pp (borderline image-use; ignores evidence entirely), and our trained v6.0-retrain verifier reaches test IMG = 70.21pp and ESG = 20.01pp. Two API detectors (Claude 3.5 Sonnet, MAIRA-2 as verifier) returned 0.0 accuracy on this run, attributable to image preprocessing mismatch between verifier-mode prompts and the MAIRA-2 processor API (and, for Claude, to per-minute rate limits on Opus-4-7). Both are pending re-runs.

5.6. Silver-label validation against PadChest-GR radiologist bboxes

PadChest-GR ships 11,995 radiologist-labeled sentences (7,037 positive-finding with bounding boxes, 4,958 explicitly-absent from the same radiologist’s reports). These constitute the only public claim-level radiologist-grounded set in chest radiology not derived from MIMIC-CXR. We attempted the paired validation in three phases. **Phase 1:** pair PadChest-GR GT sentences against the OpenI-only silver-labeled claim set — yielded zero `image_id` matches (disjoint patient-image identifiers). **Phase 2:** extract 26 PadChest-GR images from the multi-part ZIPs, run MAIRA-2 on them, decompose into 86 atomic claims, silver-label with RadFact + VERT, re-pair by `study_id` — also yielded zero matches at the validator’s Jaccard ≥ 0.30 threshold. **Phase 3:** bumped extraction budget to 500, generated 168 reports, decomposed into 565 claims — yielded 12 matches at Jaccard ≥ 0.30 from 11,995 GT sentences (0.1% of GT). Of those 12, 9 silver-labels were UNCERTAIN (excluded from agreement metrics), leaving 3 definite labels — too few for a meaningful Cohen κ (the apparent $\kappa = -0.8$ is small-sample noise). VERT additionally hit a 51% Anthropic-API error rate on this run, further reducing the usable label count. The Phase-2/3 low match-count is a vocabulary-overlap failure rather than an indexing one: MAIRA-2 on PadChest-GR images produces generic normal-pattern claims (“Heart size is normal”, “Lungs are clear”), while PadChest-GR’s positive-finding sentences are specific pathology descriptions (“At-

Table 3: Cross-site 3-way leave-one-out: image-masking gap on the held-out site. The v5.0 2-way result on OpenI $\rightarrow$ ChestX-Det10 is shown for reference.

Held-out site	IMG (pp)	ESG (pp)	Acc (held-out)	Blind?
OpenI	3.21	-0.02	0.853	yes (both)
ChestX-Det10	28.41	2.69	0.660	yes (ESG)
PadChest-GR	<i>N/A — PadChest-GR records require claim-extraction (see §5.6)</i>			
<i>2-way reference (v5.0 Tier 3): OpenI $\rightarrow$ ChestX-Det10</i>	0.29	—	0.80	yes

electasis in the right middle lung field”). The same image is described in disjoint vocabularies. Lowering the Jaccard threshold to 0.10 would yield matches but they would be on common stop-words rather than clinical content. The right validation requires either an RRG generator whose outputs cover specific findings densely (e.g., a clinical-fine-tuned variant) or a clinical-NER-based matching layer that is finding-aware; both are deferred to the camera-ready. The full pipeline is scaffolded at `v5/eval/padchest_gr_validate.py` and `v5/modal/v6_orchestrator.py:padchest_gr_rescue_chain`; supplement §7 reports the 12-match detail.

As a partial indirect signal, the ensemble Krippendorff α on the 2,707 real-RRG claims separates leakage-prone and leakage-free graders cleanly: the two MIMIC-free graders (RadFact, VERT) agree at $\alpha = 0.70$, while GREEN (MIMIC-fine-tuned) agrees with RadFact at $\alpha = 0.04$ and with VERT at $\alpha = -0.001$. This is direct evidence that GREEN’s label distribution diverges from the two MIMIC-free graders, consistent with the MIMIC-pretraining leakage hypothesis motivating our three-grader design, and argues for reporting silver-label agreement per-subset rather than aggregated.

5.7. Cross-site 3-way leave-one-out transfer

We evaluate whether the evidence-blindness mitigation transfers across sites by training three held-out-site configurations: held-out OpenI (train on ChestX-Det10 only, since PadChest-GR rows require a separate claim-extraction step documented in §6), held-out ChestX-Det10 (train on OpenI only), and held-out PadChest-GR (train on OpenI + ChestX-Det10; effectively the full 2-site configuration). All three use the v5.3-contrastive loss composition, are trained for two epochs, and have the HO filter disabled because the v5 / v6-full HO weights are positionally indexed and not transferrable across LOO train files.³ The held-out-PadChest-GR row is N/A because its test rows are raw radiologist records rather than normalized (claim, evidence, label) tuples; PadChest-GR’s role in this paper is radiologist-bbox validation of silver labels (§5.6), not cross-site training transfer.

3. An earlier version inherited HO weights across LOO files; a pre-flight review caught the row-index mismatch before launch. The LOO checkpoints therefore evaluate the consistency loss + contrastive loss contribution to cross-site transfer, not the full v5.3 pipeline. This is a conservative choice: if the IMG gap survives LOO without the HO filter, it survives the full stack.

Table 4: Evidence-blindness diagnostic on 500 real-RRG claim subset. IMG = image-masking gap; ESG = evidence-shuffling gap. Evidence-blind iff $\text{IMG} < 5\text{pp}$ or $\text{ESG} < 5\text{pp}$.

Detector	Acc	IMG (pp)	ESG (pp)	
BiomedCLIP (zero-shot)	0.538	8.0	0.0	
MedGemma-4B-IT (as verifier)	0.758	14.0	2.0	
MAIRA-2 (as verifier)	0.498	32.2	6.6	
RadFlag (MAIRA-2, $K=6$) [†]	0.664	N/A	N/A	
Claude 3.5 Sonnet (API)	—	—	—	infra-failed (anthropic)
ClaimGuard-CXR (v5.3 / v6.0-retrain) (ours)	0.911 (test)	70.21	20.01	

[†]RadFlag is a hallucination flagger, not a binary claim verifier with image / evidence inputs; the evidence-blindness counterfactuals (image-zeroing, evidence-shuffling) do not apply directly because RadFlag operates on the underlying generator’s resampled outputs. Reported accuracy is the agreement between the flag bit (flagged $\Leftrightarrow$ CONTRADICTED) and the ensemble silver labels on the 107 claims with definite SUP/CON labels (precision 0.381, recall 0.615). Full breakdown in §5.8.

5.8. Baseline landscape on real-RRG claims

We evaluate five detectors on the real-RRG claim set under the evidence-blindness diagnostic (two additional planned baselines — RadFlag replication and a ReXTrust-inspired white-box probe — are described in Appendix C):

- **CheXagent-2-3b** (Chen et al., 2024) as a zero-shot claim verifier (chat-template prompt).
- **MAIRA-2** (Bannur et al., 2024) as a zero-shot claim verifier.
- **MedGemma-4B-IT** (Google, 2025) as a zero-shot claim verifier.
- **BiomedCLIP zero-shot** (Zhang et al., 2023) with CheXzero-style caption templates.
- **ClaimGuard-CXR (v5.3 / v6.0-retrain)** (ours): the v5.3-contrastive architecture, reported on the v6 test split.

Two of the three completed binary verifier baselines (BiomedCLIP, MedGemma) are evidence-blind on the evidence-shuffling axis; MAIRA-2 (as a zero-shot claim verifier, after fixing the trust-remote-code processor handshake described in §6) is not blind on either axis ($\text{IMG} = 32.2\text{pp}$, $\text{ESG} = 6.6\text{pp}$, accuracy 0.498). The MAIRA-2 result is the strongest counter-example to the field-level-blindness hypothesis in our data: an off-the-shelf grounded RRG model used as a claim verifier passes both diagnostic gates, even though its aggregate accuracy is barely above chance. The combination matters — high IMG with low aggregate accuracy means MAIRA-2 *uses* the image but does not *verify* the claim well; aggregate accuracy and image-grounding are two different deployment-relevant axes and a verifier can fail one without failing the other. The Claude 3.5 Sonnet row is reported as infrastructure-failed rather than dropped — every one of 500 calls exhausted our rate-limit

Table 5: Per-site text-only shortcut audit (RoBERTa-base, claim+evidence input).

Site	n test	Majority acc	Text-pair acc	HO activation rate
ChestX-Det10	1,387	0.798	0.875 (+7.6pp)	68.2%
OpenI	1,587	0.854	0.914 (+6.0pp)	82.2%

retry, attributable to the anthropic 0.42 + claude-3-5-sonnet-20241022 combination on our submission Modal image; the API-baseline row will be repaired with a current Claude / Sonnet 4.6 in the camera-ready.

5.9. Text-only artifact audit

We audit the claim-only text shortcut directly by training a RoBERTa-large classifier on the claim text only (no image, no evidence), once on the raw training distribution and once on the HO-filter-downweighted distribution. On the 4,132-claim v6 test split, the pre-filter text-only model reaches 82.0% (vs majority-class 87.6%, -5.6pp); the post-filter text-only model reaches 79.5% (-8.1pp vs majority). The HO filter reduces text-only shortcut performance by 2.5 percentage points without changing the training-data row count (the filter downweights rather than drops).

A finer per-site audit using a RoBERTa-base text-pair classifier (Table 5) tells a different story. On each site separately the text-pair classifier beats the majority baseline by 6–8pp, and the HO filter would activate on 68–82% of test rows. The v6 training distribution is therefore not shortcut-free at the site level; the original text-only-vs-majority gap appears to wash out only when sites are aggregated. The HO filter’s primary contribution remains the loss-drop ablation effect (removing it alone collapses IMG from 62.9pp to 5.8pp on v5.2); the text-only-vs-majority gap should be read as confirming that the HO filter’s downweighting changes the training distribution in the direction the loss-drop ablation shows is load-bearing, not as evidence that no shortcut existed in the first place.

5.10. HO-filter activation on real-RRG hallucinations (M6 gate)

A central deployment claim of the mitigation is that the HO filter, trained on synthetic negation-flipped negatives in the v6 train split, transfers to naturally-occurring hallucinations produced by RRG models. We test this by scoring all 1,934 ensemble-resolved real-RRG silver claims (§5.5) with the HO filter and reporting the activation rate at threshold $\tau = 0.7$. Results in Table 6.

The aggregate 80.6% activation passes the M6 gate by a wide margin: the HO filter does fire on real-RRG claims, not just synthetic ones. The structure of the activation matters: the filter flags 91.6% of silver-SUPPORTED claims (most positive findings have a recognizable claim-text pattern that a text-only classifier can predict) but only 28.4% of silver-CONTRADICTED claims (real hallucinations are harder to detect from claim text alone). The filter therefore primarily downweights “easy” positive claims, leaving the model to learn from the harder positives and from the hallucinations themselves. This is the right direction for the loss the filter participates in, but a reviewer should not read the 80.6% as evidence that the filter would catch most hallucinations as a deployed standalone detector —

Table 6: HO-filter activation on the real-RRG ensemble silver-labeled claim set. Activation rate is the fraction of claims the text-pair classifier scores at ≥ 0.7 confidence on the silver label. Plan target was $\geq 40\%$; safety floor $\geq 10\%$.

Subset	n	n flagged	Activation rate
All real-RRG claims	1,934	1,559	80.6%
MedGemma-4B-IT	1,330	1,150	86.5%
MAIRA-2	305	209	68.5%
CheXagent-2-3b	299	200	66.9%
flagged silver=SUPPORTED	1,599	1,464	91.6%
flagged silver=CONTRADICTED	335	95	28.4%

Table 7: Support-score sharpness on v6 test split. Higher KS = farther from uniform, sharper distribution. Higher $p(y|x) > 0.9$ = more high-confidence predictions.

Config	Mean predicted prob	KS to uniform	Frac $p > 0.9$
v5.0-base	0.935	0.69	79.3%
v5.1-ground	0.938	0.73	82.0%
v5.2-real	0.933	0.63	77.5%
v5.3-contrast	0.929	0.62	76.9%
v6.0-retrain	0.922	0.62	76.9%

its CONTRADICTED-class detection is well below either the SUPPORTED-class detection or majority-class baselines.

5.11. Support-score sharpness across configurations

We measure the support-score (predicted-class softmax probability) distribution of each verifier on the 2,974-claim v6 test split, reporting per-configuration Kolmogorov–Smirnov distance to a uniform-on-[0.5, 1.0] reference and the fraction of predictions above 0.9 confidence. Numbers in Table 7.

The distributions are similarly peaked across configurations (mean predicted probability ≥ 0.92 for every model). The two evidence-blind configs (v5.0, v5.1) show *slightly sharper* distributions than the non-blind ones (v5.2, v5.3, v6.0). This is the opposite of what a "sharpness explains conformal coverage" hypothesis would predict, and we do not endorse the simpler version of that explanation. The right reading is that the consistency + contrastive losses make the verifier less overconfident on text-shortcut-only claims, and the inverted cfBH conformal procedure depends on the joint structure of (support score, error event), not the marginal sharpness of either alone. Per-configuration conformal FDR is reported in supplement §C alongside the joint-distribution analysis.

6. Discussion

What the table says. A training-time mitigation that targets the evidence-blindness failure mode directly — by penalizing consistent output under image-masking and by removing text-solvable examples from the training distribution — works *on the training distribution*: the image-masking gap moves from 2pp (v5.0) to 69pp (v5.3) on the in-distribution test split while aggregate accuracy stays at 92.5%. Under leave-one-site-out shift (Table 3) the same verifier regresses to evidence-blind on held-out OpenI (IMG = 3.2pp) and borderline on held-out ChestX-Det10 (IMG = 28.4pp with ESG = 2.7pp, below the ESG threshold). The diagnostic therefore also works as a deployment-shift alarm: a verifier that passes IMG on its training sites may fail the same test on a new site, and the same three counterfactual interventions that diagnose training-time blindness also detect the post-deployment version. The mitigation is a cheap lever for one class of evidence-blindness (distributional shortcuts in-distribution); cross-site generalization is a second class that it does not reach, and laterality-turning claims (IPG ≈ 0) are a third.

Why the table looks the way it does. Two pieces of the story are worth emphasizing. First, the baseline (v5.0) passes every test that an aggregate-accuracy reviewer would apply, including cross-entropy convergence and val/test agreement, while being fundamentally image-blind. That is what the NLI literature found ten years ago, reproduced here under chest radiographs. Second, the grounding loss (v5.1) produces the best *validation* accuracy of any configuration tested, while producing essentially the same image-masking gap as the cross-entropy baseline. A practitioner using validation accuracy alone as the model-selection criterion would pick the worst-grounded of our trained models.

Limitations. The benchmark is compact: approximately 7,500 images from OpenI and ChestX-Det10, yielding 25,176 labeled training claims. We ingested a third site (PadChest-GR, 4,555 radiologist-drawn sentences with bounding boxes) but did not normalize it into (claim, evidence, label) training tuples in time for this submission; the records are retained as-is in the cohort JSONLs for silver-label validation (§5.6), and a planned future step will run the same LLM-based claim-extraction pipeline we use for OpenI + ChestX-Det10 over the PadChest-GR Spanish and English reports. Until that extraction lands, the cross-site leave-one-out table in §5.7 reports two held-out configurations (OpenI and ChestX-Det10) rather than three, and the PadChest-GR held-out row is marked N/A. The per-category IMG analysis is stable at this scale but the per-pathology cells with small n are noisier than the aggregate numbers. A larger benchmark would tighten the per-category table without changing the headline claim.

The consistency loss pushes zeroed-image accuracy below 50% on v5.2/v5.3. This behavior is a consequence of the loss: if the model learns to output high confidence on **SUPPORTED** when the image is present and high confidence on **CONTRADICTED** when the image is absent, it will both clear the IMG threshold by a wide margin and produce worse-than-random masked accuracy. The right interpretation is that the model is genuinely image-dependent, not that 29.4% accuracy means something has broken.

Laterality remains partially evidence-blind across every configuration we trained. We do not present this as a failure of the mitigation; it is direct evidence that evidence-blindness is not a single bug with a single fix. Training-distribution intervention is a cheap lever that

works where it works. Where it does not, architectural changes are needed — in particular, image encoders that represent spatial arrangement at a granularity finer than the frozen BiomedCLIP ViT blocks resolve.

Finally, this paper trains on a cross-entropy + auxiliary-loss regime and evaluates on a counterfactual diagnostic. The fundamental conceptual gap — that aggregate accuracy is a weak proxy for image grounding — generalizes to other multimodal medical tasks, but the specific mitigation recipe may not. The diagnostic is domain-agnostic; the mitigation is task-specific.

7. Conclusion

Multimodal medical claim verifiers can be image-blind in ways that aggregate accuracy cannot detect. We give evidence-blindness an operational definition and show, on a public benchmark, that a cross-entropy baseline is evidence-blind, that a grounding loss alone does not fix it, and that a consistency loss combined with an adversarial hypothesis-only filter closes the image-masking gap by more than thirty times without trading off aggregate accuracy. Laterality-turning claims remain a residual blindness across every training configuration we tried. We recommend counterfactual evidence-sensitivity testing as a default check on multimodal medical AI systems before any safety-relevant deployment claim, and release the benchmark, diagnostic metrics, training code, and model weights.

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Appendix A. Training Hyperparameters

Table 8: Training hyperparameters used for all configurations.

Image backbone	BiomedCLIP ViT-B/16 (open_clip fallback)
Text backbone	RoBERTa-large
Shared dim / fusion layers	$d = 768$, 4 layers, 12 heads
Optimizer	AdamW (lr $1e-5$ encoders, $5e-5$ heads, wd 0.01)
Batch size	32 (grad accum 2)
Warmup / schedule	500-step linear warmup, cosine decay
Mixed precision	bf16
HO-filter threshold	0.7 (default)
HO-filter downweight	0.2
Consistency margin	0.2
Contrastive margin	0.2
MC-dropout samples per batch	5
Frozen image layers / text layers	8 / 16

Appendix B. Pending Experiments

The v5.4-final configuration (all five losses plus MC-dropout uncertainty regularization) was queued for training at the time of writing but blocked on Modal H100 capacity. The four configurations reported in the main text are sufficient to establish the paper’s central claim; v5.4 will be added to Table 2 in the camera-ready. Additional experiments already scaffolded (loss-component ablation, HO-threshold sweep, cross-site transfer, seven-baseline landscape) are documented in the released codebase and the camera-ready version will include their results.

Appendix C. Planned Baseline Extensions

Two baselines are scaffolded in the released codebase but were not run under the compute budget available for this submission; both will be reported in the extended version.

RadFlag replication (Zhang et al., 2024). A black-box temperature-sampling detector at `v5/eval/radflag_replica.py`. For each MAIRA-2-generated claim we sampled $K = 6$ temperature-1.0 alternative generations (paper used $K = 10$; we reduced to fit budget), decomposed each via Claude Haiku, and checked entailment via Claude Opus. On a 200-claim subset (sampled from the gated-RRG pool to match MAIRA-2 outputs), RadFlag flagged 47.5% of claims as hallucinations (95/200). Comparing the flag bit to the ensemble silver labels on the 107 claims with definite SUP/CON silver labels: precision = 0.381, recall = 0.615, accuracy = 0.664 (TP = 16, FP = 26, FN = 10, TN = 55). For comparison, the original ML4H paper reported 28% recall at 73% precision on MedVersa with $K = 10$ and 208 reports — different P/R trade-off, attributable to backbone (MAIRA-2 vs MedVersa), K (6 vs 10), and silver-label distribution. The full per-claim RadFlag JSONL is released alongside the paper code.

ReXTrust-inspired white-box probe (Hardy et al., 2025). A linear probe on generator hidden states, reimplemented from the published description (no public code or weights), scaffolded at `v5/eval/hidden_state_probe.py`. We attempted to apply the probe to MAIRA-2 in this submission; the embedding-extraction step returned an empty embedding stack on every claim under the deployed extractor implementation, which we trace to a layer-handle resolution issue specific to MAIRA-2’s trust-remote-code architecture (the model exposes hidden states only through a non-default forward path that the extractor wrapper does not invoke). The MedVersa application requires installing the `medomni` package (not pip-installable; vendored at <https://github.com/rajpurkarlab/MedVersa>). Both probe runs are deferred to the camera-ready pending an extractor refactor.

Neither omission affects the paper’s central claim. The evidence-blindness diagnostic is model-agnostic and applies to both RadFlag and ReXTrust unchanged; the expected behavior (both detectors are image-backbone-dependent and should be tested for evidence-blindness separately) is discussed in §5.8.